NaijaSense AI - Solution Paper

DSN × Bluechip Tech LLM Agent Challenge · DSAS 2026 · Team TAOTECH SOLUTIONS

Abstract

NaijaSense AI is a dual-task LLM system: **Task A** simulates a domain-aware star rating and aligned first-person review from persona + product text; **Task B** returns a grounded lifestyle recommendation **paragraph** from a persona-only narrative with mandatory **Reason-Before-Recommend** agent_reasoning. Production agents are two POST endpoints (Groq router llama-3.1-8b-instant + generator llama-3.3-70b-versatile), grounded in a single **5,011-row** Yelp/Amazon/Goodreads corpus (few-shots, silent history, and Task B stage-1 retrieval share one inverted index). A unified hub (/unified) provides demos and ablations; evaluation uses the dedicated task URLs below.

Endpoints: Task A - https://naija-sense-ai.vercel.app/task-a/user-modeling · Task B - https://naija-sense-ai.vercel.app/task-b/recommendation · Code - https://github.com/taotechs/NaijaSense-AI

1. Problem & Submission Contract

Reviews encode tone, rating bias, and domain taste. We target auditable agents - not opaque chat - with strict I/O per task.

Task	Path	Input	Output
A	POST /task-a/user-modeling	user_persona, product_details (strings)	rating, review_reasoning, review_text
B	POST /task-b/recommendation	{ user_id, persona } only	recommendations (paragraph), agent_reasoning

2. Architecture

flowchart TB C[Client] --> TA[Task A: two-pass agent] C --> TB[Task B: retrieve then rerank] TA --> R1[rating + reasoning + review] TB --> P2[parse persona + intent] P2 --> S1[Top-30 from 5k corpus] S1 --> D1[diversify + dedupe variants] D1 --> S2[Groq router rank] S2 --> S3[Groq generator paragraph] S3 --> R2[recommendations + agent_reasoning]

Task A (TaskATwoPassAgent). Parse texts → infer product domain (food/tech/etc.) from product details only → **Pass 1 (router):** JSON rating + rationale with domain few-shots → **Pass 2 (generator):** 2-4 sentence review with **rating locked**. Heuristic fallback if JSON fails.

Task B (TaskBPipelineAgent). Parse persona and intent (lifestyle, advisory-only learning/career, or team-culture hiring) → **Stage 1:** top-30 candidates from the full **5,011-row** corpus index (diversified, variant titles collapsed) → **Stage 2 (Groq):** router ranks item_ids → generator writes one grounded recommendations paragraph with agent_reasoning; stage-1 template fallback if Groq fails.

Demo hub (not scored). /unified adds intent routing, silent history by user_id, critique→regenerate, streaming NDJSON, and pidgin/Yoruba modes - used for UX and scripts/run_real_benchmark.py ablations only.

3. Data, Models & Evaluation

Corpus: review_corpus.jsonl **- 5,011** rows (Yelp 2,498, Amazon 2,482, Goodreads 31 + curated seeds), fully indexed for Task A and Task B. **Models:** Groq router + generator via GROQ_API_KEY and ORCHESTRATOR_PROVIDER=groq. **Metrics:** Task A - ROUGE-1/L, token-F1 (BERTScore fallback), RMSE; Task B - NDCG@10, Hit@10 (evals.py, scripts/run_real_benchmark.py).

Task A ablations (orchestrator path, N=20): full ROUGE-1 0.161, RMSE 1.25; no_llm ROUGE-1 0.126 - LLM is the main lever. Submission adds **two-pass rating lock** to cut score-text drift. **Task B ablations** (legacy deterministic ranker, N=25): Hit@10 0.20 vs ~0.50 random on hard same-domain distractors; **submission** uses Groq two-step rerank over the stage-1 pool to address this.

4. Reproducibility & Limitations

git clone https://github.com/taotechs/NaijaSense-AI.git && cd NaijaSense-AI cp .env.example .env   # GROQ_API_KEY docker compose up --build python scripts/smoke_api.py http://127.0.0.1:8000

Swagger: /docs · Smoke tests cover both task endpoints. **Limits:** (1) Ablations ≠ submission pipelines - score /task-a and /task-b directly; (2) Task A persona must be self-contained (no silent history on submission route); (3) Task B quality depends on stage-1 recall and LLM grounding; (4) Task B is lifestyle recommendation, not HR or general career consulting - advisory-only personas are steered to books/tech picks, not random restaurants.

5. Conclusion

NaijaSense AI ships two deployable, explainable agents from one repo: Task A with domain-aware two-pass review simulation, Task B with corpus retrieval, Groq rerank, and mandatory agent_reasoning in a single recommendation paragraph. One solution paper covers both tasks; each task has its own agent URL. The hub supports iteration; the endpoints in §1 are the primary evaluation surface.